1.

1.1.

2022 pp. 40 - 41

1.2.

2022 pp. 43, 47 - 54

12 $X\mathrm{cm}^2$ Y g

(1) $Y = aX + b$

(2)

(3)

$$\begin{aligned}(X, Y) = (11, 23),\quad (23, 49),\quad (5, 10),\quad (50, 115),\quad (32, 64),\quad (21, 45)\\ (16, 35),\quad (39, 84),\quad (25, 50),\quad (45, 100),\quad (5, 11),\quad (34, 70)\end{aligned}$$

2.

2.1.

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(1)

(1)

(2) (2) (2)

(1) (2)

(3) 1 6 (2)

1 2 3 4 5 6 (2) $\Omega = \{1, 2, 3, 4, 5, 6\}$ (3)

(2) Ω (1)

(3) (2)

2.2.

2022 2022 2022 pp. 78, 80, 83

(1)

(2)

(3)

(4) 3

2.3.

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1 X

c

$P(X = x) = c \times x \quad (x = 1, 2, 3, 4, 5, 6)$ $\times$

(1) c

(2) $x = 1, 2, 3, 4, 5, 6$ $P(X = x)$

(3) X $E[X]$

(4) $P(X \geq 4)$

2.4.

2023 pp. 81, 86

$$f(x) = \begin{cases} \dfrac{c}{x^3} (x \geq 10) \\ 0 \quad (x < 10) \end{cases}$$

(1) c

(2)

(4)

(5)

(6) 1

2.8.

2025

(1) X μ σ^2

(2) 200 60 6 42
78

3.

3.1.

2024 p. 107

6 4 1 2

(1) 2

(2) 2

(1) $\dfrac{6}{10} \cdot \dfrac{5}{9} = \dfrac{1}{3}$

(2) $\dfrac{6}{10} \cdot \dfrac{5}{9} + \dfrac{4}{10} \cdot \dfrac{6}{9} = \dfrac{3}{5}$

3.2.

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3 5

(1) 1 1

(a) 2

(b) 2

(c) 2 1

(2) 2

1

(3) (1) (2)

(1) (a) $\dfrac{3}{8} \cdot \dfrac{2}{7} = \dfrac{6}{56} = \dfrac{3}{28}$

(b) $\dfrac{3}{8} \cdot \dfrac{2}{7} + \dfrac{5}{8} \cdot \dfrac{3}{7} = \dfrac{21}{56} = \dfrac{3}{8}$

(c) $\dfrac{{}^{6}/_{56}}{{}^{21}/_{56}} = \dfrac{6}{21} = \dfrac{2}{7}$

(2) $\dfrac{3}{8}$

(3) 1

3.3.

2022 pp. 107 - 108

1 2 3 4 2

X_1 X_2 $Y = |X_1 - X_2|$

(1) Y

(2) Y

(3) Y

(1)

Y	0	1	2	3	
P	$1/4$	$3/8$	$1/4$	$1/8$	1

(2) $E[Y] = \frac{1}{4}\cdot 0 + \frac{3}{8}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 = \frac{5}{4}$

(3) $\mathrm{Var}[Y] = E[Y^2] - (E[Y])^2 = \left(\frac{1}{4}\cdot 0^2 + \frac{3}{8}\cdot 1^2 + \frac{1}{4}\cdot 2^2 + \frac{1}{8}\cdot 3^2\right) - \left(\frac{5}{4}\right)^2 = \frac{5}{2} - \frac{25}{16} = \frac{15}{16}$

3.4.

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(X, Y)

$$f(x,y) = \begin{cases} 6e^{-2x-3y} (x \geq 0, y \geq 0) \\ 0 \qquad\qquad (\text{otherwise}) \end{cases}$$

X Y

$x \geq 0$ $y \geq 0$

$$f_1(x) = \int_{-\infty}^{\infty} f(x,y)dy = \int_0^{\infty} 6e^{-2x-3y}dy = 6e^{-2x}\left[-\frac{1}{3}e^{-3y}\right]_0^{\infty} = 2e^{-2x}$$

$$f_2(y) = \int_{-\infty}^{\infty} f(x,y)dx = \int_0^{\infty} 6e^{-2x-3y}dx = 6e^{-3y}\left[-\frac{1}{2}e^{-2x}\right]_0^{\infty} = 3e^{-3y}$$

$$f_1(x) = \begin{cases} 2e^{-2x} & (x \geq 0) \\ 0 & (x < 0) \end{cases} \qquad f_2(y) = \begin{cases} 3e^{-3y} & (y \geq 0) \\ 0 & (y < 0) \end{cases}$$

$x \geq 0$ $y \geq 0$

$$f(x,y) = f_1(x) f_2(y)$$

X Y

3.5.

2026

$X \quad Y$

$$\int_0^\infty e^{-x}dx = 1, \qquad \int_0^\infty xe^{-x}dx = 1$$

(1) $f(x,y) = \begin{cases} xe^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$

(a) $f_X(x) \quad f_Y(y)$

(b) $X \quad Y$

(2) $f(x,y) = \begin{cases} c(x+y)e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$

(a) c

(b) $X \quad Y$

(1) (a) $x \geq 0 \quad y \geq 0$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = xe^{-x}\underbrace{\int_0^\infty e^{-x}dx}_{=1} = xe^{-x}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = e^{-y}\underbrace{\int_0^\infty xe^{-x}dx}_{=1} = e^{-y}$$

$$f_X(x) = \begin{cases} xe^{-x} & (x \geq 0) \\ 0 & (\text{otherwise}) \end{cases}, \qquad f_Y(y) = \begin{cases} e^{-y} & (y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

(b) $f(x,y) = xe^{-x}e^{-y} = f_X(x)f_Y(y)$ $X \quad Y$

(2) (a) $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy = 1$

$$\int_0^\infty\int_0^\infty c(x+y)e^{-x-y}dxdy = c\underbrace{\int_0^\infty xe^{-x}dx}_{=1}\underbrace{\int_0^\infty e^{-y}dy}_{=1} + c\underbrace{\int_0^\infty e^{-x}dx}_{=1}\underbrace{\int_0^\infty ye^{-y}dy}_{=1} = 2c$$

$$c = \frac{1}{2}$$

(b) $x \geq 0, y \geq 0$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \frac{1}{2}xe^{-x}\underbrace{\int_0^\infty e^{-y}dy}_{=1} + \frac{1}{2}e^{-x}\underbrace{\int_0^\infty ye^{-y}dy}_{=1} = \frac{1}{2}(x+1)e^{-x}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \frac{1}{2}e^{-y}\int_0^{\infty}$$

3 $\frac{1}{2}$

(1) 1 2

(2) 1 2 A 2 3 B A B

(3) $P(A|B)$

(1) $\Omega = (M,F)^3$ (M,M,F) 1 2 3 $\equiv$ Ω “M ” $\equiv$ $\Omega_0 = [(F,F,F)]_\equiv$ $\Omega_1 = [(M,F,F)]_\equiv$ $\Omega_2 = [(M,M,F)]_\equiv$ $\Omega_3 = [(M,M,M)]_\equiv$ $|\Omega_0| = \binom{3}{0} = 1$ $|\Omega_1| = \binom{3}{1} = 3$ $|\Omega_2| = \binom{3}{2} = 3$ $|\Omega_3| = \binom{3}{3} = 1$ $|\Omega| = 8$

$\frac{4/8}{7/8} = \frac{4}{7}$

(2) A $A = \{(M,M,F),(M,M,M)\}$ B $B = \{(F,M,M),(M,M,M)\}$ $A \cap B$ $A \cap B = \{(M,M,M)\}$ $|\Omega|$ $P(A) = \frac{1}{4}$ $P(B) = \frac{1}{4}$ $P(A \cap B) = \frac{1}{8}$ A B $P(A \cap B) = P(A)P(B)$ A B

(3) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/8}{1/4} = \frac{1}{2}$

3.9.

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2 A 4 B 2 7 C 2 9

(1) A B

(2) A C

X 2 Y A $X = 4$ B $X + Y = 7$ C $X + Y = 9$

(1)
- $P(A) = \frac{1}{6}$

- $P(B) = P(X+Y=7) = \frac{1}{6}$
- $P(A \cap B) = P(X=4, X+Y=7) = P(X=4, Y=3) = \frac{1}{36}$

$P(A \cap B) = P(A)P(B)$ A B

(2)
- $P(A) = \frac{1}{6}$
- $P(C) = P(X+Y=9) = \frac{1}{9}$
- $P(A \cap C) = P(X=4, X+Y=9) = P(X=4, Y=5) = \frac{1}{36}$

$P(A \cap C) \neq P(A)P(C)$ A C

3.10.

2023 pp. 72 - 73, 109 - 110, 114 - 115

2 E F $P(E) = a$ $P(F) = b$

(1) $P(E \cup F)$ a b

(2) $P(E \cap F^c | E \cup F)$ a b F^c F

(1) $P(E \cup F) = a + b - ab$

(2) $P(E \cap F^c | E \cup F) = \dfrac{P(E \cap F^c)}{P(E \cup F)} = \dfrac{a - ab}{a + b - ab}$

3.11.

2025

1 5

X Y $Z = X - Y$

(1) X Y

(2) Z

(3) Z

(4) Z

3.12.

2025 pp. 118 - 119

(1) 2 A B

(2) 500 1 X X

2 K_1 K_2 $3:1$

K_1,K_2 0.98 0.90

0.01

K_1

3.13.

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3 5 1

2

(1) 1

(2) 2 1

3.14.

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1 10% 3

(1)

(2) 2

(1) $\dfrac{9}{10}\cdot\dfrac{9}{10}\cdot\dfrac{9}{10} = 0.7290$

(2) $\dfrac{\frac{9}{10}\cdot\frac{1}{10}}{\frac{1}{10}+\frac{9}{10}\cdot\frac{1}{10}+\frac{9}{10}\cdot\frac{9}{10}\cdot\frac{1}{10}} = \dfrac{90}{271} \approx 0.3321$

3.15.

2022 pp. 118 - 119

5 4 3

1 100 g 45 g 30 g

(1) 1

40 g

(2) 2

100 g 2

(3) 3

150 g 3

(1) $\dfrac{\binom{5}{1}}{\binom{12}{1}-\binom{3}{1}} = \dfrac{5}{9}$

(2) $\dfrac{\binom{5}{2}}{\binom{12}{2}-\binom{7}{2}} = \dfrac{10}{66-21} = \dfrac{2}{9}$

(3) $\dfrac{\binom{5}{3}}{\binom{12}{3}-\binom{7}{3}} = \dfrac{10}{220-35} = \dfrac{10}{185} = \dfrac{2}{37}$

3.16.

2022 2024 pp. 118 - 119

A 0.1%

A /

99% 10%

(1)

(2) A 1

A A B 2

C

(1) $P(A|B) = \dfrac{P(A)P(B|A)}{P(A)P(B|A)+P(A^c)P(B|A^c)} = \dfrac{0.001 \cdot 0.99}{0.001 \cdot 0.99 + 0.999 \cdot 0.10} = \dfrac{99}{10089} \approx 0.0098$

(2) $P(A|C) = \dfrac{P(A)P(C|A)}{P(A)P(C|A)+P(A^c)P(C|A^c)} = \dfrac{0.001 \cdot 0.99^2}{0.001 \cdot 0.99^2 + 0.999 \cdot 0.10^2} = \dfrac{9801}{109701} \approx$ 0.0893

3.17.

2023 pp. 118 - 119

99%

99% 0.5%

1

(1)

(2)

(3)

(4)

A B

(1) $P(A \cap B) = P(A)P(B|A) = 0.005 \cdot 0.99 = 0.00495$

(2) $P(A^c \cap B) = P(A^c)P(B|A^c) = 0.995 \cdot 0.01 = 0.00995$

(3) $P(B) = P(A \cap B) + P(A^c \cap B) = 0.00495 + 0.00995 = 0.0149$

$$P(A^c|B) = \frac{P(A^c \cap B)}{P(B)} = \frac{0.00995}{0.0149} \approx 0.6678$$

(4) $P(A|B^c) = \dfrac{P(A \cap B^c)}{P(B^c)} = \dfrac{P(A)P(B^c|A)}{1 - P(B)} = \dfrac{0.005 \cdot 0.01}{0.9851} \approx 0.0001$

3.18.

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S S

K

- S K 99% 1%
- S K 99% 1%

(1) K S

(2) (1)

(1) S A K B

$$P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)} = \frac{\frac{1}{10000} \cdot \frac{99}{100}}{\frac{1}{10000} \cdot \frac{99}{100} + \frac{9999}{10000} \cdot \frac{1}{100}} = \frac{99}{10098} \approx 0.0098$$

(2) K

S

3.19.

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$\boldsymbol{A}$

$\boldsymbol{B}$

(1) $P(A|B) = 0.98$ $P(A|B^c) = 0.005$ $P(B) = 0.001$

(2) $P(A|B^c) = 0.003$ $P(B) = 0.0005$ $P(B|A) \geq 0.142$ $P(A|B)$

(1)

$$P(B|A) = \frac{P(B)P(A|B)}{P(B)P(A|B) + P(B^c)P(A|B^c)} = \frac{0.001 \cdot 0.98}{0.001 \cdot 0.98 + 0.999 \cdot 0.005} = \frac{980}{5975} \approx 0.1640$$

(2) (1) $$P(B|A) = \frac{0.0005 \cdot P(A|B)}{0.0005 \cdot P(A|B) + 0.9995 \cdot 0.003} \geq 0.142$$

$$P(A|B) \geq \frac{1}{0.0005} \cdot \frac{1}{1 - 0.142} + 0.142 \cdot 0.9995 \cdot 0.003 \approx 0.9925$$

3.20.

pp. 118 - 119

500 1 XXX 2

K_1 K_2 $K_1 : K_2 = 3 : 1$

$\boldsymbol{K_1}$ 0.98

$\boldsymbol{K_2}$ 0.90

0.01

(1) K_1

(2) (1)

K_1 K_2 A_1 A_2

A_0 B

(1) $P(A_1) = \dfrac{1}{500} \cdot \dfrac{3}{4} = \dfrac{3}{2000}$ $P(A_2) = \dfrac{1}{500} \cdot \dfrac{1}{4} = \dfrac{1}{2000}$ $P(A_0) = \dfrac{499}{500} = \dfrac{1996}{2000}$

$$P(B) = \underbrace{\frac{3}{2000} \cdot \frac{98}{100}}_{P(A_1 \cap B)} + \underbrace{\frac{1}{2000} \cdot \frac{90}{100}}_{P(A_2 \cap B)} + \underbrace{\frac{1996}{2000} \cdot \frac{1}{100}}_{P(A_0 \cap B)} = \frac{294 + 90 + 1996}{200000} = \frac{2380}{200000}$$

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{294}{2380} \approx 0.1235$$

(2)

3.21.

2023 pp. 122 - 124

p $0 < p < 1$ X Y

$$f(k) = \begin{cases} p & (k = 1) \\ 1 - p & (k = -1) \end{cases}$$

$Z = XY$

(1) $E[Z]$

(2) X Z $E[(X - E[X])(Z - E[Z])]$

(3) X Z p p Y Z

(4) (3) p $\Pr[X + Y + Z \leq 2]$

(1) $E[X] = E[Y] = p - (1 - p) = 2p - 1$ $E[Z] = E[XY] = E[X]E[Y] = (2p - 1)^2$

(2) $E[XZ] = E[X^2 Y] = E[Y]$

$$\mathrm{Cov}(X, Z) = E[XZ] - E[X]E[Z] = (2p - 1)[1 - (2p - 1)^2] = 4p(1 - p)(2p - 1)$$

(3) $\mathrm{Cov}(X, Z) = 0$ ____ $0 < p < 1$ $p = \dfrac{1}{2}$

$P(X = a, Y = b) = P(X = a)P(Y = b) = \dfrac{1}{4}$

(4) $(X, Y) = (1, 1)$ $X + Y + Z = 3 > 2$

$$\Pr[X + Y + Z \leq 2] = 1 - \Pr[X + Y + Z > 2] = \frac{3}{4}$$

3.22.

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2 X Y

(1) a

(2) X Y $\mathrm{Cov}(X, Y)$

(3) X Y

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

(1) $8a = 1$ $a = \dfrac{1}{8}$

(2) X Y

- $E[X] = \dfrac{3}{8} \cdot 1 + \dfrac{2}{8} \cdot 2 + \dfrac{3}{8} \cdot 3 = 2$
- $E[Y] = \dfrac{2}{8} \cdot 1 + \dfrac{3}{8} \cdot 10 + \dfrac{3}{8} \cdot 100 = \dfrac{83}{2}$
- $E[XY] = \dfrac{664}{8} = 83$

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 83 - 2 \cdot \frac{83}{2} = 0$$

(3) $P(X = 2) = \dfrac{1}{4}$ $P(Y = 1) = \dfrac{1}{4}$ $P(X = 2, Y = 1) = 0$

$$P(X = 2, Y = 1) \neq P(X = 2)P(Y = 1)$$

X Y

3.23. n

2023 p. 127

n 1 n $(n + 1)$

$$\binom{n}{1} \cdot \underbrace{\left(\frac{1}{6}\right)}_{2\times 1} \cdot \underbrace{\left(\frac{1}{6}\right)^{n-1}}_{1\times(n-1)} = \frac{n}{6^n}$$

4.

4.1.

2025

2 $B(2,p)$

$X \sim B(2,p)$ $P(X=k)=\binom{2}{k}p^k(1-p)^{2-k}$

$$\begin{aligned}
E[X] &= \sum_{k=0}^{2} kP(X=k) = P(X=1)+2P(X=2) \\
&= \binom{2}{1}p(1-p)+2\cdot\binom{2}{2}p^2 \\
&= p[2(1-p)+2p] = 2p
\end{aligned}$$

$$\begin{aligned}
\mathrm{Var}(X) &= E[(X-E[X])^2] = E[(X-2p)^2] \\
&= \sum_{k=0}^{2}(k-2p)^2P(X=k) \\
&= (0-2p)^2P(X=0)+(1-2p)^2P(X=1)+(2-2p)^2P(X=2) \\
&= 4p^2\binom{2}{0}(1-p)^2+(1-2p)^2\binom{2}{1}p(1-p)+4(1-p)^2\binom{2}{2}p^2 \\
&= 8p^2(1-p)^2+2p(1-p)(1-2p)^2 = 2p(1-p)
\end{aligned}$$

4.2.

2024 p. 141

1 6

(1) 10 5

(2) 100 25

(1) X X $Bin\left(10,\frac{1}{6}\right)$

$$\begin{aligned}
P(X\geq 5) &= \sum_{k=5}^{10}P(X=k) = \sum_{k=5}^{10}\binom{10}{k}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{10-k} = \frac{1}{6^{10}}\sum_{k'=0}^{5}\binom{10}{k'}5^{k'} \\
&= \frac{1}{6^{10}}(1\cdot 1+10\cdot 5+45\cdot 25+120\cdot 125+210\cdot 625+252\cdot 3125) \\
&= \frac{934926}{60466176} = 0.01546\ldots \approx 0.0155
\end{aligned}$$

(2) X X $Bin\left(100, \dfrac{1}{6}\right)$ $E[X] = \dfrac{50}{3}$ $\sigma[X] = \dfrac{5\sqrt{5}}{3}$
$n(1-p) > np = \dfrac{50}{3} > 5$ n –
X $N\left(100, \left(5\sqrt{5}/3\right)^2\right)$

$$P(X \geq 25) \approx P(X > 24.5) = P\left(Z > \frac{24.5 - {}^{50}\!/_3}{{}^{5\sqrt{5}}\!/_3}\right) = P\left(Z > \frac{23.5}{5\sqrt{5}}\right)$$
$$= P\left(Z > 0.94\sqrt{5}\right) \approx P(Z > 2.102) \approx 1 - 0.9821 = 0.0179$$

4.3.

2023 2026 pp. 93 - 95, 141

p n
X 2 $\mathrm{Bin}(n,p)$ $q = \left(\dfrac{X}{n}\right)$

(1) $P(|q-p| \leq 0.02) \geq 0.95$ n

(2) (1) n (2)
(3) p

(3) (1) -
n

(4) (2) (3) n

(1) q $|q-p| \leq 0.02$ 0.02
$P(|q-p| \leq 0.02) \geq 0.95$ 0.02
0.95

(2) $\mathrm{Var}(q) = \dfrac{p(1-p)}{n}$ $\lambda = \dfrac{0.02}{\sqrt{\mathrm{Var}(q)}}$

$$P(|q-p| \leq 0.02) = P\left(|q-p| < \lambda\sqrt{\mathrm{Var}(q)}\right) \geq 1 - \frac{1}{\lambda^2} = 1 - \frac{p(1-p)}{0.02^2 n} \geq 0.95$$

$\dfrac{p(1-p)}{0.02^2 n} \leq 0.05$ $n \geq 50{,}000p(1-p)$
$0^2 < p(1-p) \leq \left(\dfrac{1}{2}\right)^2$ $n \geq 12{,}500$

(3) – n q
$N(p, \mathrm{Var}(q))$

$$P(|q-p| \le 0.02) = P\left(\left|\frac{q-p}{\sqrt{\mathrm{Var}(q)}}\right| \le \frac{0.02}{\sqrt{\mathrm{Var}(q)}}\right) = P\left(|Z| \le \frac{0.02}{\sqrt{\mathrm{Var}(q)}}\right) \ge 0.95$$

$$P(|Z| \le 1.96) \approx 0.95$$

$$\frac{0.02}{\sqrt{\mathrm{Var}(q)}} \ge 1.96$$

$$n \ge \left(\frac{1.96}{0.02}\right)^2 p(1-p) = 9,604p(1-p) \ge 2,401$$

(4)

n

n

$$P(|q-p| \le \varepsilon) \ge 1-\alpha$$

$$\sigma = \sqrt{\frac{p(1-p)}{n}}$$

- ٮ

$$P(|q-p| \le \varepsilon) = P\Big(|q-p| < \frac{\varepsilon}{\sigma} \cdot$$

α

$\ge$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$y = f(x)$ (x, y)

(2) /

(3)

(1) $f' = -\frac{(x-\mu)}{\sigma^2} f \quad f'' = \frac{(x-\mu)-\sigma}{\sigma} \cdot \frac{(x-\mu)+\sigma}{\sigma} f$

(2) $f(x)$

(3)

0.5

4.5.

2022 pp. 121 - 122, 143

(1) X Y $N(170, 6^2)$

(a) $W = \dfrac{X+Y}{2}$

(b) X Y W

(2) X Y

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \qquad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$W = XY$

(1) (a) $E[W] = E\left[\frac{1}{2}X + \frac{1}{2}Y\right] = \frac{1}{2}E[X] + \frac{1}{2}E[Y] = \frac{1}{2} \cdot 170 + \frac{1}{2} \cdot 170 = 170$

(b) $\mathrm{Var}(W) = \mathrm{Var}\left(\frac{1}{2}X + \frac{1}{2}Y\right) = \left(\frac{1}{2}\right)^2 \mathrm{Var}(X) + \left(\frac{1}{2}\right)^2 \mathrm{Var}(Y) = \frac{1}{4} \cdot 6^2 + \frac{1}{4} \cdot 6^2 = 18$

(2)

$$E[X] = \int_0^{2\pi} x f(x) dx = \frac{1}{2\pi} \int_0^{2\pi} x dx = \frac{1}{4\pi} [x^2]_0^{2\pi} = \frac{(2\pi)^2}{2\pi} = \pi$$

$$E[Y] = \int_{-3}^{9} y g(y) dy = \frac{1}{12} \int_{-3}^{9} y dy = \frac{1}{24} [y^2]_{-3}^{9} = \frac{1}{24}(9^2 - (-3)^2) = 3$$

X Y

$$E[W] = E[XY] = \pi \cdot 3 = 3\pi$$

4.6.

2024 pp. 147 - 151

(1) X $N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma)$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$

(2) (

)

(3)

(a) $z(0.01)$

(b) $\Phi(z_0) = 0.01$ z_0

(c) $z(0.005)$

(1) $P(\mu + a\sigma \leq X \leq \mu + b\sigma) = \Phi(b) - \Phi(a)$ $\Phi(a) - \Phi(-a) = 2\Phi(a) - 1$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma) = \Phi(1) - \Phi(-1) = 0.6826$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = \Phi(2) - \Phi(-2) = 0.9544$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = \Phi(3) - \Phi(-3) = 0.9974$

(2) 3 $[\mu - 3\sigma, \mu + 3\sigma]$ 3σ

(3) (a) 2.327

(b) -2.327

(c) 2.575

4.7.

2024 pp. 149 - 151

(1) a Z $N(0,1)$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2) X $N(m, \sigma^2)$

$$P\left(|X-m| \geq \frac{\sigma}{4}\right)$$

4

(3) m σ n

$\overline{X}$

$$P\left(|\overline{X}-m| \geq \frac{\sigma}{4}\right) \leq 0.02$$

n

(1) $f(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$ $f(-x) = f(x)$

$$P(|Z| \geq a) = \int_{-\infty}^{-a} f(x)dx + \int_{a}^{\infty} f(x)dx = \int_{a}^{\infty} f(x)dx + \int_{a}^{\infty} f(x)dx = 2P(Z \geq a)$$

(2) $P\left(|X-m| \geq \frac{\sigma}{4}\right) = P\left(|Z| \geq \frac{1}{4}\right) \approx 0.8026 \approx 0.803$

(3) $E\left[\overline{X}\right] = m$ $\sigma\left[\overline{X}\right] = \frac{\sigma}{\sqrt{n}}$

$$P\left(|\overline{X}-m| \geq \frac{\sigma}{4}\right) = P\left(|Z| \geq \frac{\sqrt{n}}{4}\right) \leq 0.02$$

$$P(Z \leq 2.33) = 0.9901 \qquad P(|Z| > 2.33) \approx 0.0198 \approx 0.02$$

$$\frac{\sqrt{n}}{4} \geq 2.33 \Longrightarrow n \geq (2.33 \cdot 4)^2 = 86.56$$

$n = 87$

4.8.

2024 2026 pp. 149 - 150

4

E 350,198 66.31

23.55 X $N(66.31, (23.55)^2)$

(1) $P(X \geq 80)$

(2) (3-a)

(3) 80 95

(4) 20%

(1) $n = 350,198$ $\mu = 66.31$ $\sigma = 23.55$ X $N(\mu, \sigma^2)$ $Z = \dfrac{X-\mu}{\sigma}$ $N(0,1)$ $P(X \geq 80) = P\Big(Z \geq \dfrac{80-\mu}{\sigma}\Big) \approx P(Z \geq 0.58)$ $P(X = 80) \approx 0$ $P(X \geq 80) = P(Z \geq 0.58) = 1 - \Phi(0.58)$ $\Phi(0.58) \approx 0.7910$

$$P(X \geq 80) \approx 1 - 0.7910 = 0.2090$$

(2) E 1

80 E 80

(3) X $[80, 90]$ $P(80 \leq X \leq 90)$

$$\begin{aligned} P(80 \leq X \leq 90) &= P\Big(Z \leq \frac{90-\mu}{\sigma}\Big) - P\Big(Z \leq \frac{80-\mu}{\sigma}\Big) \\ &= P\Big(Z \leq \frac{90-66.31}{23.55}\Big) - P\Big(Z \leq \frac{80-66.31}{23.55}\Big) \\ &= \Phi(1.01) - \Phi(0.58) \approx 0.8438 - 0.7190 = 0.1248 \end{aligned}$$

$$nP(80 \leq X \leq 90) \approx 350,198 \times 0.1248 = 43,705$$

(4) 20% 80% φ $P(X \leq \varphi) = 0.80$ $P(X \leq \varphi) = P\Big(Z \leq \dfrac{\varphi-\mu}{\sigma}\Big) = \Phi\Big(\dfrac{\varphi-\mu}{\sigma}\Big)$ $\Phi(0.84) \approx 0.7955 \fallingdotseq 0.80$ $\Phi\Big(\dfrac{\varphi-\mu}{\sigma}\Big) \approx \Phi(0.84)$ $\dfrac{\varphi-\mu}{\sigma} \approx 0.84$ $\varphi \approx \mu + 0.84\sigma = 66.31 + 0.84 \times 23.55 = 86.0920 \approx 86$ 86

4.9.

2025 pp. 149 - 150

4

120

10 100 15

C

(1) 95% C

(2)

(1) $X \sim N(120, 10^2)$ $P(X \geq C) = \Phi\Big(-\dfrac{C-120}{10}\Big) = 0.95\ (= \Phi(1.65))$

$$-\frac{C-120}{10} \approx 1.65 \Longrightarrow C \approx 120 - 1.65 \times 10 = 103.5$$

(2) $Y \sim N(100, 15^2)$ $P(Y \geq C) = 1 - \Phi\left(\frac{103.5-100}{15}\right) \approx 1 - 0.5910 = 0.4090$

4.10.

2022 pp. 153 - 154

(1) 6

(2) 10000

(3) 10000 4900 5100

(1) $\binom{6}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^3 = \frac{20}{2^6} = \frac{5}{16}$

(2) X $Bin\left(10000, \frac{1}{2}\right)$

$$\begin{cases} E[X] = 10000 \times \frac{1}{2} = 5000 \\ \sigma[X] = \sqrt{10000 \times \frac{1}{2} \times \frac{1}{2}} = 50 \end{cases}$$

(3) $np = n(1-p) = 5000$ – X $N(5000, 50^2)$

$P(4900 \leq X \leq 5100) \approx P(4899.5 < X < 5100.5) = P(-2.01 < Z < 2.01) \approx 0.9556$

4.11.

pp. 93 - 95, 153 - 154

200 60 6

X

(1) $42 \leq X \leq 78$

(2)

(3) X $N(60, 6^2)$ $P(42 \leq X \leq 78)$

$n = 200$ $\mu = 60$ $\sigma = 6$

(1)

$$P(42 \leq X \leq 78) = P(|X - 60| < 18) = P(|X - \mu| < 3\sigma) \geq 1 - \frac{1}{3^2} = \frac{8}{9}$$

(2) $Y = nX$

$$Y = nP(42 \leq X \leq 78) \geq n \times \frac{8}{9} = 177.7...$$

178

(3) $X \sim N(\mu, \sigma^2)$

$$P(42 \leq X \leq 78) = P(-3 \leq Z \leq 3) = 0.9974$$

5.

5.1.

2023 2025 pp. 168 - 169

2

$Bin(n,p)$ $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$ np $n(1-p)$

– $N(np, np(1-p))$

n p $\binom{n}{k} p^k (1-p)^{n-k} \to e^{-\lambda}\frac{\lambda^k}{k!}$

5.2.

2024 pp. 93, 168 - 169

1 ATM ATM 1 40

ATM 1 60

ATM

/

60

75

90

105

120

$M/M/1$ λ μ

$\rho = \frac{\lambda}{\mu}$ $\rho < 1$ $L = \frac{\rho}{1-\rho}$ $W = \lambda^{-1} L = \frac{\mu^{-1}}{1-\rho}$

$W = \frac{1/90}{1 - 60/90}\ \mathrm{h} = \frac{1}{30}\ \mathrm{h} = 120\ \mathrm{s}$

5.3.

2023 pp. 166 - 169

(1) 3 5

(2) 1 2 1 4

(1) 5 X $X \sim Bin(5, 0.3)$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.7)^5 = 1 - 0.16807 = 0.83193$$

(2) 1 X $X \sim Po(2)$

$$P(X \geq 4) = 1 - P(0 \leq X \leq 3) = 1 - e^{-2}\left(1 + 2 + 2 + \frac{4}{3}\right) = 1 - \frac{19}{3}e^{-2}$$

$$\approx 1 - \frac{19}{3} \cdot \frac{18}{133} = \frac{57}{399} \approx 0.1429$$

$$e^2 = 1 + \cfrac{2}{(1-1) + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + \cfrac{1}{9 + \ddots}}}}} \approx 1 + \cfrac{2}{0 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + 0}}}} = \frac{18}{399} \tag{5.1}$$

5.4.

2025

4

(1) 1 3

2

(2) X $P(X = 3) = 5P(X = 5)$

$P(X \leq 3)$

(1) $X \sim Po(3)$

$$P(X \geq 2) = 1 - P(0 \leq X \leq 1) = 1 - e^{-3}(1 + 3) = 1 - \frac{4}{e^3}$$

$$\approx 1 - 4 \cdot \frac{221}{4439} = \frac{3555}{4439} \approx 0.8009$$

$$e^3 \approx 1 + \cfrac{6}{(2-3) + \cfrac{9}{6 + \cfrac{9}{10 + \cfrac{9}{14 + \cfrac{9}{18+0}}}}} = \frac{4439}{221} \tag{5.2}$$

(2) $P(X=3) = e^{-\lambda}\dfrac{\lambda^3}{3!} = e^{-\lambda}\dfrac{\lambda^3}{6}$ $\quad 5P(X=5) = 5e^{-\lambda}\dfrac{\lambda^5}{5!} = e^{-\lambda}\dfrac{\lambda^3}{6}\left(\dfrac{\lambda}{2}\right)^2$ $\quad \lambda = 2$

$X \sim Po(2)$

$$P(X \leq 3) = e^{-2}\left(1 + 2 + 2 + \frac{4}{3}\right) \approx \frac{18}{133} \cdot \frac{19}{3} = \frac{342}{399} \approx 0.8751$$

(5.1)

5.5.

2023 pp. 166 - 169

1% 200

(1) X

(2)

(3) 3

(2) (3)

(1) $P(X=k) = \dbinom{n}{k}\left(\dfrac{1}{100}\right)^k\left(\dfrac{99}{100}\right)^{200-k} \approx e^{-2}\dfrac{2^k}{k!}$

(2) $P(X=0) \approx e^{-2} \approx \dfrac{18}{133} \approx 0.1353$

(3) $P(X \geq 3) = 1 - P(0 \leq X \leq 2) = 1 - e^{-2}(1+2+2) \approx 1 - 5 \cdot \dfrac{18}{133} = \dfrac{43}{133} \approx 0.3233$

5.6.

2024 pp. 153 - 154, 166 - 169

4

(1) 240 6 38 45

(2) 0.1% 500 3

(1) $n = 240$ $p = \dfrac{1}{6}$ 6 X $Bin(n,p)$ $np =$ 40 $n(1-p) = 200$ 5 –

X $N(np, np(1-p)) = N\left(40, \dfrac{100}{3}\right)$

$$\begin{aligned}
P(38 \le X \le 45) \approx P(37.5 \le X \le 45.5) &= P\left(\frac{37.5-40}{{}^{10}\!/\!\sqrt{3}} \le Z \le \frac{45.5-40}{{}^{10}\!/\!\sqrt{3}}\right) \\
&= P\left(-0.25\sqrt{3} \le Z \le 0.55\sqrt{3}\right) \approx P(-0.43 \le Z \le 0.95) \\
&= 0.8289 - 0.3336 = 0.4953
\end{aligned}$$

(2) $n = 500$ $p = \dfrac{1}{1000}$ X $Bin(n,p)$ n P X $Po(\lambda)$

$\lambda = np = \dfrac{1}{2}$

$$\begin{aligned}
P(X \ge 3) = 1 - P(0 \le X \le 2) &= 1 - e^{-\frac{1}{2}}\left(1 + \frac{1}{2} + \frac{1}{8}\right) = 1 - e^{-\frac{1}{2}}\frac{13}{8} \\
&\approx 1 - \frac{12 \cdot 2^2 - 6 \cdot 2 + 1}{12 \cdot 2^2 + 6 \cdot 2 + 1} \cdot \frac{13}{8} = 1 - \frac{37}{61} \cdot \frac{13}{8} = \frac{7}{488} \approx 0.0143
\end{aligned}$$

$$\sqrt[n]{e} \approx 1 + \cfrac{2}{(2n-1) + \cfrac{1}{6n+0}} = \frac{12n^2 + 6n + 1}{12n^2 - 6n + 1} \tag{5.3}$$

5.7.

2026

$p = 0.001$ $n = 500$

X

$$e^{-0.5} \approx 0.6065, \qquad \sqrt{0.5} \approx 0.7071$$

(1) (a) X

(b) $E[X]$ $\mathrm{Var}(X)$

(2)

(a)

(b) $P(X \ge 3)$

(3)

(a)

(b) $P(X \geq 3)$

(4)

(a) (2) (3)

(b)

np

(1) (a) X $Bin(500, 0.001)$

(b) $E[X] = np = 0.5$ $\mathrm{Var}(X) = np(1-p) = 0.4995$

(2) (a) n p X $Po(0.5)$

(b) $P(X \geq 3) = 1 - P(0 \leq X \leq 2) = 1 - e^{-0.5}(1 + 0.5 + 0.125) = 1 - e^{-0.5} \cdot 1.625 \approx 1 - 0.6065 \cdot 1.625 \approx 0.01444$

(3) (a) $E[X] = np = 0.5$ $\mathrm{Var}(X) = np(1-p) = 0.4995$

(b) $P(X \geq 3) \approx P(X \geq 2.5) = P\left(Z > \dfrac{2.5 - 0.5}{\sqrt{0.4995}}\right) \approx P\left(Z > \dfrac{2}{\sqrt{0.5}}\right) \approx P(Z > 2.83) \approx 0.0023$

(4) (a)

(b) $np = 0.5$ X

5.8.

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p $0 < p < 1$ $(1-p)$ N $N > 1$

(1) $N = 5$

(2) t $0 \leq t \leq N$

(3) m $m = N$ $0 \leq m \leq N$ m

(4) n $0 \leq n \leq N$ $N - n$ $P(n \& N, p)$

(5) (4) n

(6) n $\lambda = Np$ $N \to \infty$ (4) $P(n\&N, p)$ $P(n;\lambda)$ $\lim_{N\to\infty}\left(1+\frac{x}{N}\right)^N = e^x$

(1) $p \cdot p \cdot (1-p) \cdot p \cdot (1-p) = p^3(1-p)^2$

(2) p^t

(3) $E[m] = \sum_{m=0}^{N-1} p^m(1-p)\cdot m + p^N \cdot m = \frac{p}{1-p}(1-p^N)$

(4) $P(n\&N,p) = \binom{N}{n} p^n(1-p)^{N-n}$

(5) $E[n] = \sum_{n=0}^{N} nP(n\&N,p) = Np$

(6) $\lambda = Np$

$$
\begin{aligned}
&\lim_{N\to\infty}\binom{N}{n}p^n(1-p)^{N-n}\\
&= \lim_{N\to\infty}\frac{N!}{n!(N-n)!}p^n(1-p)^{N-n}\\
&= \lim_{N\to\infty}\frac{1}{n!}N(N-1)(N-2)\cdots(N-n+1)\left(\frac{\lambda}{N}\right)^n\left(1-\frac{\lambda}{N}\right)^{N-n}\\
&= \lim_{N\to\infty}\frac{\lambda^n}{n!}\left(1-\frac{1}{N}\right)\left(1-\frac{2}{N}\right)\cdots\left(1-\frac{n-1}{N}\right)\left(1-\frac{\lambda}{N}\right)^{-n}\left(1+\frac{-\lambda}{N}\right)^N\\
&= \frac{\lambda^n}{n!}e^{-\lambda}
\end{aligned}
$$

6.

6.1.

2022 p. 174

2

(1)

(2)

(1)

(2)

6.2.

2025

μ σ^2 n $X_1, X_2, ..., X_n$.

(1) $X_1, X_2, ..., X_n$

(2) .

(3) n $S_n = X_1 + X_2 + \cdots + X_n$

.

(4)

(1)

(2) $\forall \varepsilon > 0 : \lim_{n\to\infty} P\left(\left|\overline{X} - \mu\right| < \varepsilon\right) = 1$

(3) $\overline{X} = \dfrac{S_n}{n}$ n $N\left(\mu, \dfrac{\sigma^2}{n}\right)$

$$S_n = n\overline{X} \quad N\left(n\mu, n^2\frac{\sigma^2}{n}\right) = N(n\mu, n\sigma^2)$$

(4)

6.3.

2025 p. 179

1180 20

(1) 25 1170

(2) N 0.9 1175

N

(1) 25 $X_1, X_2, \ldots, X_{25}$ 1180 20^2

$\overline{X}$ $N\left(1180, \dfrac{20^2}{25}\right)$

$$P\left(\overline{X} > 1170\right) = P\left(Z > \frac{1170 - 1180}{4}\right) = P(Z > -2.5) = 0.9938$$

(2) N $\overline{X}$ $N\left(1180, \dfrac{20^2}{N}\right)$

$$P\left(\overline{X} > 1175\right) = P\left(Z > \frac{1175 - 1180}{20/\sqrt{N}}\right) = P\left(Z > -\frac{\sqrt{N}}{4}\right)$$

$$P(Z > -1.29) \approx 0.9015 > 0.9$$

$-1.29 \geq -\dfrac{\sqrt{N}}{4}$ $\sqrt{N} \geq 1.29 \times 4 = 5.16$ $N \geq (5.16)^2 = 26.6256$

$N \geq 27$

6.4.

2025

1.2 kg 0.1 kg

100 122.5 kg

$X_1, \ldots, X_{100}$ $X = X_1 + \cdots + X_{100}$

$E[X] = 100 \times 1.2$ kg $= 120$ kg $\sigma[X] = \sqrt{100} \times 0.1$ kg $= 1$ kg

X $N(120 \text{ kg}, 1 \text{ kg})$

(P

\

6.5.

2022 pp. 202 - 205

μ σ^2 3 X_1 X_2 X_3

(1) .

(a) $T_1 = X_1 + X_2 - X_3$

(b) $T_2 = \dfrac{X_1 + 2X_2 + X_3}{4}$

(c) $T_3 = \dfrac{X_1 + X_2 + X_3}{3}$

μ

(2) $\{T_1, T_2, T_3\}$

(1) μ

(a) $E(T_1) = E(X_1) + E(X_2) - E(X_3) = \mu + \mu - \mu = \mu$

(b) $E(T_2) = \frac{1}{4}E(X_1) + \frac{2}{4}E(X_2) + \frac{1}{4}E(X_3) = \frac{1}{4}\mu + \frac{2}{4}\mu + \frac{1}{4}\mu = \mu$

(c) $E(T_3) = \frac{1}{3}E(X_1) + \frac{1}{3}E(X_2) + \frac{1}{3}E(X_3) = \frac{1}{3}\mu + \frac{1}{3}\mu + \frac{1}{3}\mu = \mu$

(2) T_3

(a) $V(T_1) = 1^2\sigma^2 + 1^2\sigma^2 + (-1)^2\sigma^2 = 3\sigma^2$

(b) $V(T_2) = \left(\frac{1}{4}\right)^2\sigma^2 + \left(\frac{2}{4}\right)^2\sigma^2 + \left(\frac{1}{4}\right)^2\sigma^2 = \frac{3}{8}\sigma^2$

(c) $V(T_3) = \left(\frac{1}{3}\right)^2\sigma^2 + \left(\frac{1}{3}\right)^2\sigma^2 + \left(\frac{1}{3}\right)^2\sigma^2 = \frac{1}{3}\sigma^2$

6.6.

2023 2023 2025 pp. 176 - 180, 202 - 205

(1) 0 1.2 3

$\{-1, 0, 1\}$

(a)

(b)

(c)

(2)

(3)

(4) θ $\Theta_n = T(X_1, X_2, ..., X_n)$ 3

(a)

(b)

(c)

(5) 3

(1) (a) 1.2

(b) $\frac{1}{3}[(-1)^2 + 0^2 + 1^2] = \frac{2}{3}$

(c) $\frac{3}{2} \times \frac{2}{3} = 1$

(2)

(3)

(4) (a) $\hat{\theta}_n$ θ

$$E\left[\hat{\theta}_n\right] = \theta$$

(b) n $\hat{\theta}_n$ θ

$$\forall \varepsilon > 0 : \lim_{n \to \infty} P\left(\left|\hat{\theta}_n - \theta\right| < \varepsilon\right) = 1$$

(c) $\hat{\theta}_n$ $\forall \hat{\theta}'_n : V\left(\hat{\theta}_n\right) \leq V\left(\hat{\theta}'_n\right)$

(5) n $n \to \infty$

6.7.

2022 2024 p. 205

1000 10

32, 86, 78, 49, 48, 77, 97, 49, 51, 74

1000

$\overline{X}$

$$\overline{X} = \frac{1}{10}(32 + 86 + 78 + 49 + 48 + 77 + 97 + 49 + 51 + 74) = 64.1$$

$\overline{X^2}$

$$\overline{X^2} = \frac{1}{10}(32^2 + 86^2 + 78^2 + 49^2 + 48^2 + 77^2 + 97^2 + 49^2 + 51^2 + 74^2) = 4502.5$$

σ^2

$$\sigma^2 = \overline{X^2} - \left(\overline{X}\right)^2 = 4502.5 - (64.1)^2 = 4502.5 - 4108.81 = 393.69$$

U^2

$$U^2 = \frac{10}{9}\sigma^2 = \frac{10}{9} \times 393.69 \approx 437.43$$

6.8.

2025 pp. 179, 216

p n

X

(1) X .

(2) n X

(3) p_0 95%

n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq 114)$

(1) X $Bin(n, p)$

(2) – X n

$N(np, np(1-p))$

(3) $2z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n}}$ n

$$z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n'}} = \frac{1}{2}z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n}}$$

$$n' \quad n' = \frac{1}{4}n$$

(4) $np = 1900 \cdot 0.05 = 95$ $np(1-p) = 1900 \cdot 0.05 \cdot 0.95 = 90.25 = 9.5^2$ X

$N(95, 9.5^2)$

$$P(76 \leq X \leq 114) \approx P(75.5 < X < 114.5) = P\left(\frac{75.5-95}{9.5} < Z < \frac{114.5-95}{9.5}\right)$$
$$\approx P(-2.05 < Z < 2.05) \approx 0.9596$$

6.9.

2022 pp. 180 - 182

p p 10,000

3,500 $p = \frac{3500}{10000} = 0.35$

(1) p

(2) p 2

(1) X $N(np, np(1-p))$ p 95%

$$\left[0.35 - 1.96\sqrt{\frac{0.35 \cdot 0.65}{10000}}, 0.35 + 1.96\sqrt{\frac{0.35 \cdot 0.65}{10000}}\right] \approx [0.3407, 0.3593]$$

p 0.35

(2) $10,000 \times 2^2 = 40,000$

6.10.

2025

(1) n A Y

$q = \frac{Y}{n}$ A p $100(1-\alpha)\%$

(2) S TV T

140 32 T

(a) p 95%

(b) p 3% 0.95

(c)

6.11.

2024 2024 pp. 223 - 225

(1) $Bin(1, p)$ $X_1, X_2, ..., X_n$

(a)

(b)

(2) n A Y

(a) $q = \dfrac{Y}{n}$ A p $100(1-\alpha)\%$

(b)

(3) S TV T

140 32 T

(a) p 95%

(b) p 3% 0.95

(c)

6.12.

2024 pp. 223 - 225

(1) p 0.8 0.9 0.99
$\hat{p}$ p 0.02

(2) p 0.95 $\hat{p}$
p 0.03

6.13.

2025

2 p S $1-p$ F
m S k $B(m;p)$

(1) $B(m;p)$ $f(k)$.

(2) X $B(m;p)$ X .

(3) p $l(p)$.

(4) p .

(5) . .

(a) (4) ?

(b) (4) ?

(c) (4) ?

6.14.

2025

$N(\mu, \sigma^2)$ μ σ^2

(1) μ

(2) σ^2

6.15.

2025

(1)

(2)
/ σ^2 n
N

(3) /

7.

① 仮説の検定

② 統計量の検定

③ 優位水準と棄却域の設定

④ 帰無仮説の棄却・選択

7.1.

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(1)

(2)

(3) (4)

5% (3)

5% (5)

(1)

(1)

(6) (6) (7)

- (8:)
- (9:)

7.2.

2022 pp. 227 - 234

(1)

(2)

7.3.

2023 pp. 2235 - 236

8000

25 7700

5%

640

7.4.

2023 pp. 238 - 240

T

S 365 6

330, 350, 385, 321, 340, 365

5%

7.5.

2025 pp. 238 - 240

T

S 360 6

350, 355, 385, 390, 370, 365

(1) 10%

(2) 5%

7.6.

2023 2024 2024 2025 pp. 235 - 236

(1) μ $N(\mu, 49)$ 49

$H_0 : \mu = 4,$ $H_1 : \mu > 4$

m

(a) 0.05 $m = 6$ 2

(b) 0.01 $m = 7$ 2

(2) $H_1 : \mu = 5$ 0.05

2 0.2 0.1

7.7.

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(1) 2 A B 2

10 12

A 2.04 0.03 B

2.02 0.02 2

5%

2

(2) A B

5%

7.8.

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100 70

68 15

5%

7.9.

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3 1

(1) A 10 A

“3 1 ” A

5%

(2) 10 7 “3 1 ”

7.10.

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5 1 5%

7.11.

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10 X p

$H_0 : p = \frac{1}{2}$ H_0

(1) X

(2) X

(a) $H_1 : p < \frac{1}{2}$ 5%

(b) $H_1 : p \neq \frac{1}{2}$ 2%

7.12.

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5

1 p

(1) 5 5 0

(a) $p = \frac{1}{2}$

(b) $p = \frac{2}{3}$

(2) $p \geq \frac{1}{2}$

(a) 3

$p = \frac{1}{2}$

(b) $p = \frac{3}{4}$ “ 3

”

7.13. 2

256 116 5%

.

(1) 2

(2) 2 10%

7.14.

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p $0 < p < 1$

$H_0 : p = \dfrac{1}{2}$ “ 3 3

3 H_0 H_0 ”

(1) 1 1

(2) 2 2

(3) 2 $\dfrac{1}{3}$ p

7.15.

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3 A B C A 3 B 2 C 1 1

1

(1) A A $\dfrac{1}{16}$ B B $\dfrac{1}{9}$ C C $\dfrac{1}{25}$

1

(2) 960 640

(1) 5%